Alexandru-Georgian Oprea

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Research Interests

Mathematical physics, particularly the application of algebraic, categorical, homological and homotopical methods to theoretical physics. Specific interests include algebraic topology, homotopy theory, higher algebra, category theory, operads, Hopf algebras, quantum groups and Grothendieck–Teichmüller theory, especially their connections to quantum field theory and related mathematical structures.

Additional interests include pure mathematics, quantum information theory and the mathematical foundations of theoretical physics.

Education

ETH Zürich - Department of Physics *Zürich, Switzerland*

MSc Physics *Sep 2025–Mar 2027*

- Focus on theoretical physics and advanced mathematics, with coursework in algebraic topology, algebraic geometry, representation theory, differential geometry and mathematical physics.
- Current GPA: 5.57/6; 121 ECTS completed (65 towards MSc degree)

University of Groningen - Faculty of Science and Engineering *Groningen, Netherlands*

BSc Astronomy *Sep 2022–Aug 2025*

- Graduated summa cum laude; completed 320 ECTS
- GPA: 9.1/10; ranked 2nd of 115 students

University of Groningen *Groningen, Netherlands*

Honours College - Selective interdisciplinary programme for high-performing students. *Apr 2023–Aug 2025*

Research Experience and Publications

Free Subalgebras of the Grothendieck–Teichmüller Lie Algebra (MSc Thesis) *ETH Zürich - Department of Physics*

Supervisor Prof. Thomas H. Willwacher *Sep 2026–Present*

- Investigating the algebraic structure of the Grothendieck–Teichmüller Lie algebra, focusing on the construction and characterization of free Lie subalgebras and their generators.
- Studying the underlying structures through operad theory, braid groups, Hopf algebras, quantum groups and Tannaka–Krein reconstruction.
- Exploring connections between Grothendieck–Teichmüller theory, algebraic structures arising in mathematical physics and related questions in quantum field theory.

Alternatives to Dark Matter (MSc Research Project) *ETH Zürich - Department of Physics*

Supervisor Dr. A. Lazopoulos *Spring 2026*

- Investigated theoretical alternatives to the ΛCDM dark matter paradigm, focusing on observational challenges and the theoretical motivations for modified gravity.
- Focused on conformal (Weyl) gravity, studying its fourth-order field equations, spherically symmetric solutions, and implications for galactic rotation curves and cosmology, alongside MOND and its relativistic extension TeVeS.
- Examined theoretical consistency issues, including conformal invariance, ghost degrees of freedom, unitarity, and Ostrogradsky instabilities.

Short Introduction to Vertex Operator Algebras (Seminar Paper) *ETH Zürich - Department of Physics*

Supervisor Dr. Christoph A. Keller *Jan 2026*

- Developed an introduction to vertex algebras and vertex operator algebras (VOAs) as the algebraic framework underlying two-dimensional conformal field theories, covering mutually local fields, normal ordering, and the state–field correspondence.
- Constructed the Heisenberg vertex algebra and its associated VOA, describing the rank-one free boson CFT and its Virasoro algebra representation.
- Examined how operator product expansions (OPEs), conformal symmetry, and VOA representation theory enable the computation of CFT correlation functions.

Milky Way CGM Temperature Asymmetry (Publication) *University of Groningen - Kapteyn Astronomical Institute*

Supervisors Prof. Filippo Fraternali and Prof. Else Starkenburg ***Published*** *Mar 2026*

- **Oprea, A.** et al. (2026). "Temperature asymmetry in the Milky Way's hot circumgalactic medium induced by the Magellanic Clouds." **Monthly Notices of the Royal Astronomical Society**, 547(4). DOI: 10.1093/mnras/stag319

Hierarchical Topology of the Cosmic Web Tidal Force Field (BSc Thesis) *University of Groningen - Kapteyn Astronomical Institute*

Supervisors Prof. Rien van de Weijgaert (Kapteyn Institute) and Prof. Marcello Seri (Bernoulli Institute) *Mar 2025–Jun 2025*

- Applied persistent homology, Morse theory and topological data analysis to characterize the hierarchical structure of the Cosmic Web in the Illustris-3-Dark simulation.
- Developed and applied topological descriptors, including (p)-norms, persistence entropy and Wasserstein distances, to quantify structural stability and evolution.

Symmetries in the Wigner Phase-Space Description of the One-Dimensional Quantum Harmonic Oscillator (Honours project) *University of Groningen - Van Swinderen Institute*

Supervisor Prof. Daniël Boer (Van Swinderen Institute) *Feb 2024–Apr 2024*

- Studied the Wigner phase-space formulation of the quantum harmonic oscillator using quasiprobability distributions.
- Investigated the action of symplectic transformations and the associated $SU(1,1)$ symmetry structure.

Selected Advanced Coursework

University of Groningen / ETH Zürich *Groningen & Zürich*

- **Pure Mathematics:** Algebraic Topology I–II (I: 5.00/6, II: 5.75/6), Algebraic Geometry (5.00/6), Commutative Algebra (5.25/6), Representation Theory (5.25/6), Introduction to Lie Groups (5.75/6), Differential Geometry II (6.00/6), Analysis on Manifolds, Functional Analysis, Probability and Measure
- **Geometry and Mathematical Physics:** Integrable Systems and Symplectic Geometry (5.50/6), Symmetric Spaces (6.00/6), Quantum Field Theory I–II (I: 5.50/6, II: 5.00/6), Introduction to Conformal Field Theory, General Relativity (5.50/6)
- **Other Theoretical Physics:** Quantum Information Processing I–II

Work Experience

Calculus I, Calculus II, Electricity and Magnetism *University of Groningen*

Teaching Assistant *Sep 2023–Feb 2025*

- Conducted two weekly tutorials, explaining course material and guiding students through problem-solving exercises.
- Graded homework, midterms and final examinations.

Fundamentals of Electronics *University of Groningen*

Teaching Lab Assistant *Feb 2025–Mar 2025*

- Supervised laboratory sessions for groups of up to 20 students, covering circuit simulation, operational amplifiers and advanced electronics.
- Assisted students with LTspice simulations and assessed laboratory reports.

Awards and Certificates

Jul 2026	**Academic Scholarship,** Prahova Community Foundation	*Romania*
Nov 2023	**KHMW Young Talent Incentive Awards,** Royal Holland Society of Sciences and Humanities	*Netherlands*
Feb 2021	**Mention,** The Mathematical Gazette Olympiad - National stage	*Romania*
2018&2019	**Mention and Silver Medal,** Physics Olympiad - National stage	*Romania*

Technical Skills

Programming	Python, C/C++, Java, C#
Scientific Software	Mathematica, LTspice, NumPy, SciPy, Matplotlib, pandas, PyTorch
Tools	Git, LaTeX, BibTeX, Zotero, Docker
Operating Systems	Linux, macOS, Windows

Volunteer experience

Astronomy Outreach Volunteer *Groningen, Netherlands*

Kapteyn Astronomical Institute *Oct 2022–Oct 2023*

- Organized public stargazing activities, operated Schmidt–Cassegrain telescopes and delivered introductory talks on astronomy and observational techniques.

Languages

Romanian	Native
English	C1
French	Working proficiency

References available upon request.